2011-CE
MATH
PAPER 1

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG CERTIFICATE OF EDUCATION EXAMINATION 2011

MATHEMATICS PAPER 1

Question-Answer Book

8.30 am – 10.30 am (2 hours)

This paper must be answered in English

Please stick the barcode label here.

Candidate Number							

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7, 9 and 11.

2. This paper consists of THREE sections, A(1), A(2) and B. Each section carries 33 marks.

3. Attempt ALL questions in Sections A(1) and A(2), and any THREE questions in Section B. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

4. Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.

5. Unless otherwise specified, all working must be clearly shown.

6. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.

7. The diagrams in this paper are not necessarily drawn to scale.

8. No extra time will be given to candidates for sticking the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

©香港考試及評核局 保留版權
Hong Kong Examinations and Assessment Authority
All Rights Reserved 2011

FORMULAS FOR REFERENCE

SPHERE	Surface area	=	$4\pi r^2$
	Volume	=	$\frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	=	$2\pi rh$
	Volume	=	$\pi r^2 h$
CONE	Area of curved surface	=	πrl
	Volume	=	$\frac{1}{3}\pi r^2 h$
PRISM	Volume	=	base area × height
PYRAMID	Volume	=	$\frac{1}{3}$ × base area × height

Answers written on this page will not be marked.

Please stick the barcode label here.

SECTION A(1) (33 marks)

Answer ALL questions in this section and write your answers in the spaces provided.

1. Make k the subject of the formula $\frac{mk-t}{k+t}=4$. (3 marks)

2. Simplify $\frac{x^{65}}{(x^4y^3)^2}$ and express your answer with positive indices. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

3. Factorize

(a) $81m^2 - n^2$,

(b) $81m^2 - n^2 + 18m - 2n$.

(3 marks)

4. (a) Round off 8091.1908 to the nearest ten.

(b) Round up 8091.1908 to 3 significant figures.

(c) Round down 8091.1908 to 3 decimal places.

(3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Please stick the barcode label here.

5. The following table shows the numbers of books read by some students in a certain month.

Number of books read	6	9	12	15
Number of students	4	7	10	x

If the mean number of books read by these students is 11, find the value of x. (3 marks)

6. In a summer camp, the ratio of the number of boys to the number of girls is $7:6$. If 17 boys and 4 girls leave the summer camp, then the number of boys and the number of girls are the same. Find the original number of girls in the summer camp. (4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

7. The marked price of a birthday cake is 360 . The birthday cake is sold at a discount of 45% on its marked price.

(a) Find the selling price of the birthday cake.

(b) If the marked price of the birthday cake is 80% above its cost, determine whether there will be a gain or a loss after selling the birthday cake. Explain your answer.

(4 marks)

8. The coordinates of the point A are $(-4, 6)$. A is rotated anticlockwise about the origin O through $90°$ to B . M is the mid-point of AB .

(a) Find the coordinates of M .

(b) Is OM perpendicular to AB ? Explain your answer.

(5 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Please stick the barcode label here.

9. In Figure 1, AD is the angle bisector of $\angle BAC$. It is given that $\angle ABD = \angle ACD$.

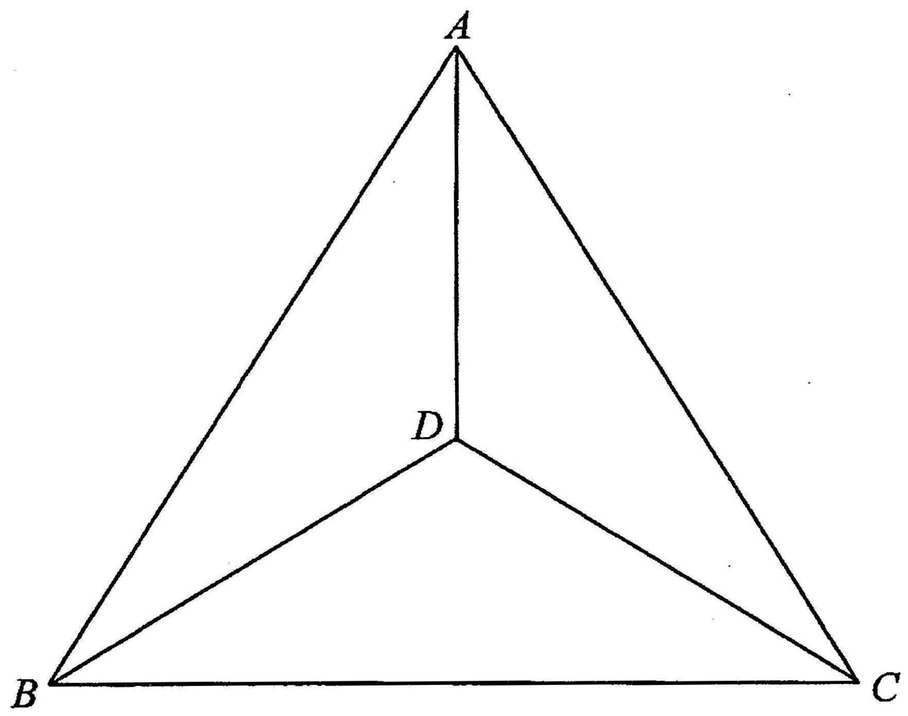


Figure 1

(a) Prove that $\Delta ABD \cong \Delta ACD$.

(b) If $\angle BAD = 31°$ and $\angle ACD = 17°$, find $\angle CBD$.

(5 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

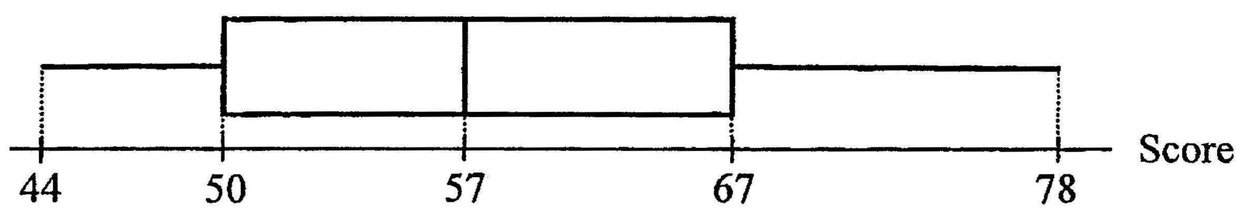
SECTION A(2) (33 marks)
Answer ALL questions in this section and write your answers in the spaces provided.

10. The student union of a school conducts two surveys to measure the extent of the students' satisfaction on the services provided by the school library. A score from 0 to 100 is used to measure the extent of satisfaction on the services, with 0 indicating absolute dissatisfaction and 100 indicating absolute satisfaction. The stem-and-leaf diagram below shows the distribution of scores rated by 32 students in the first survey.

Stem (tens)	Leaf (units)
2	3 3
3	2 4 6 6
4	2 3 3 5 5 7
5	1 6 6 7 7 8 8 8 8
6	3 3 5 5 6 6 7 7 9 9
7	5

(a) Find the median, the range and the inter-quartile range of the above distribution. (3 marks)

(b) After six months, the student union conducts the second survey to these 32 students. The box-and-whisker diagram below shows the distribution of scores rated by these students in the second survey.



(i) Is the distribution of scores in the second survey less dispersed than the first survey? Explain your answer.

(ii) The chairman of the student union claims that at least 25% of these students have a greater extent of satisfaction shown in the second survey than the first survey. Do you agree? Explain your answer.

(4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Please stick the barcode label here.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

11. It is given that $f(x)$ is the sum of two parts, one part varies as x^2 and the other part varies as x. Suppose that $f(-2)=28$ and $f(6)=-36$.

(a) Find $f(x)$. (3 marks)

(b) Figure 2 shows the graph of $y=3(x-6)^2+k$ and the graph of $y=f(x)$, where k is a constant. The two graphs have the same vertex.

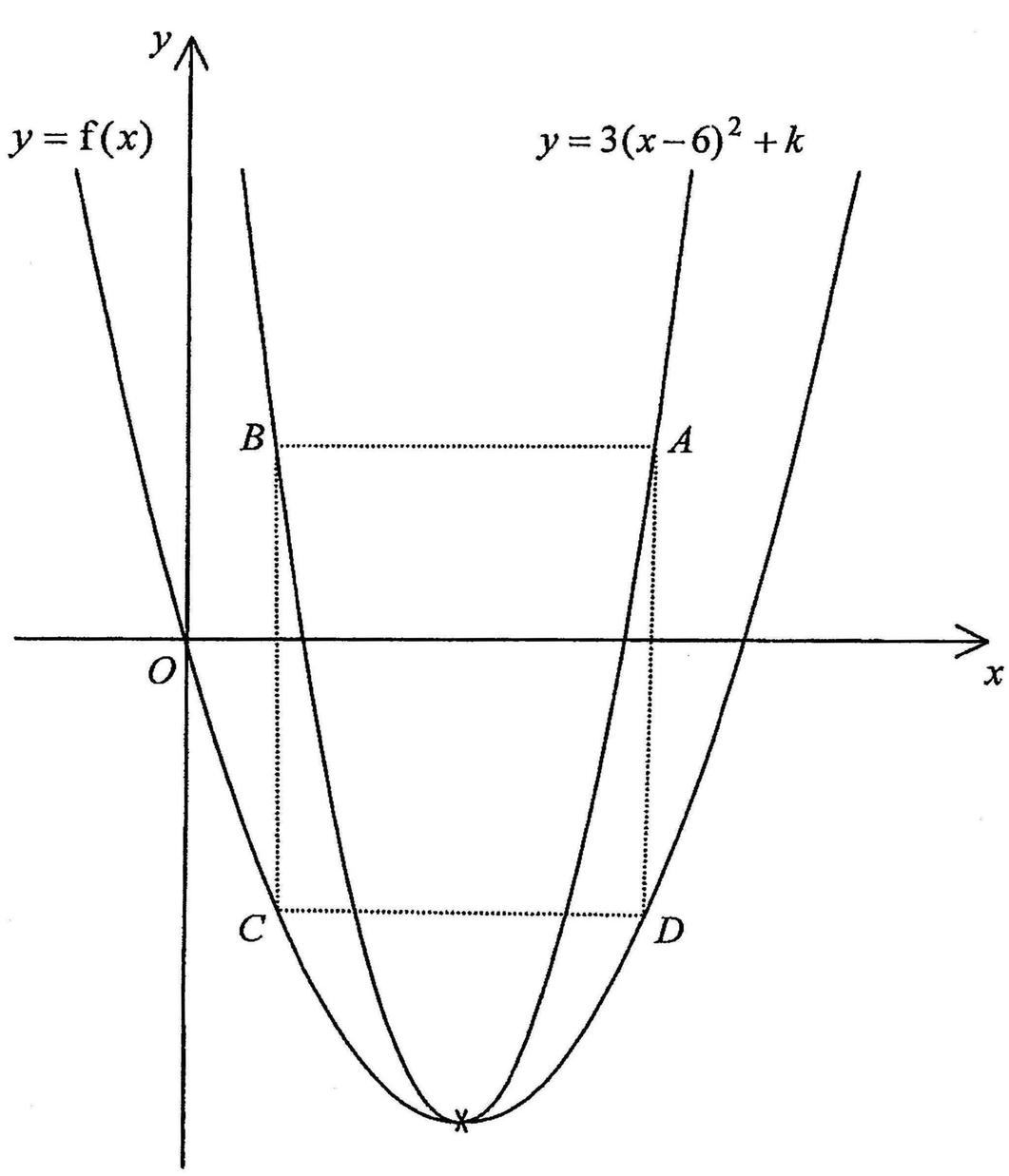


Figure 2

(i) Find the value of k.

(ii) It is given that A and B are points lying on the graph of $y=3(x-6)^2+k$ while C and D are points lying on the graph of $y=f(x)$. Also, $ABCD$ is a rectangle and AB is parallel to the x-axis. The x-coordinate of A is 10. Find the area of the rectangle $ABCD$.

(5 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Please stick the barcode label here.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

12. In Figure 3, $ABCD$ is a trapezium, where AB is parallel to CD. P is a point lying on BC such that $BP = x$ cm. It is given that $AB = 3$ cm, $BC = 11$ cm, $CD = k$ cm and $\angle ABP = \angle APD = 90°$.

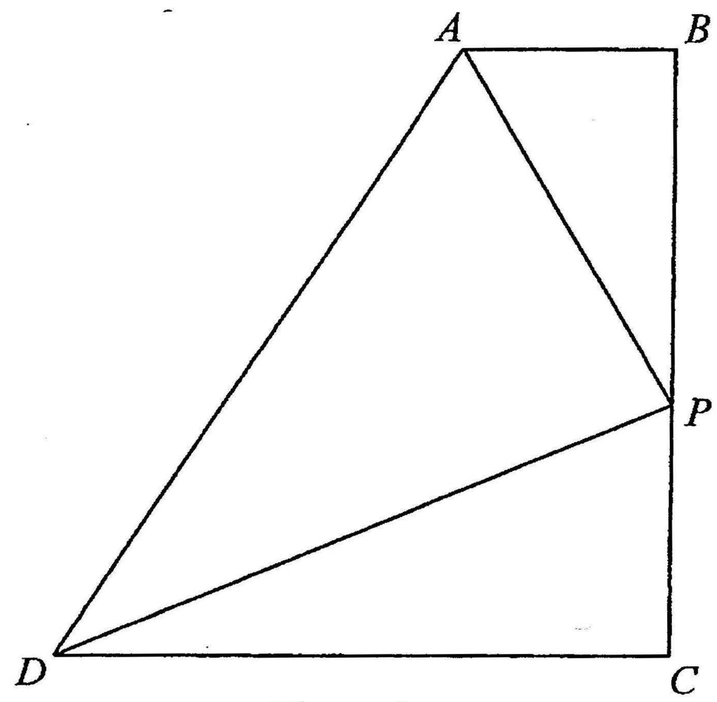


Figure 3

(a) Prove that $\Delta ABP \sim \Delta PCD$. (3 marks)

(b) Prove that $x^2 - 11x + 3k = 0$. (2 marks)

(c) If k is an integer, find the greatest value of k. (4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

13. Figure 4(a) shows the thin paper sector $OXYZ$ of area $2880\pi\,\text{mm}^2$. By joining OX and OZ together, $OXYZ$ is folded to form an inverted right circular conical container as shown in Figure 4(b).

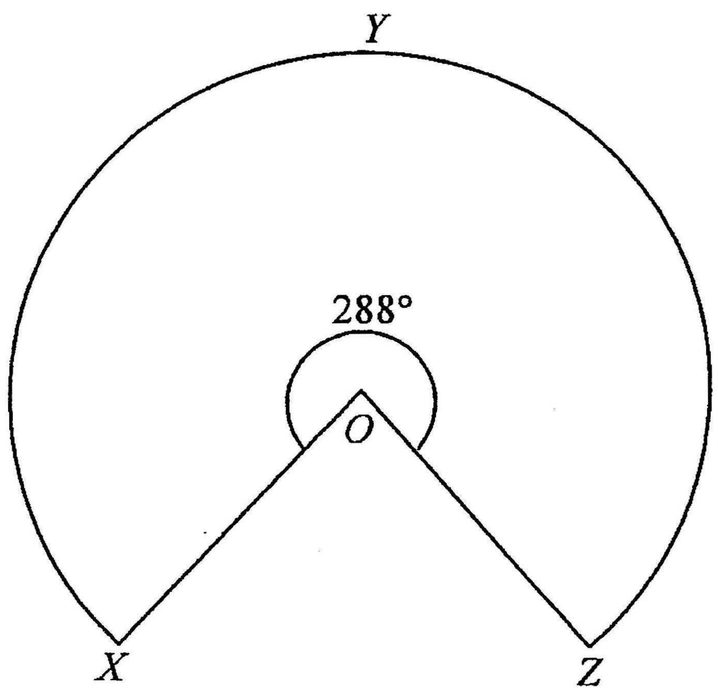


Figure 4(a)

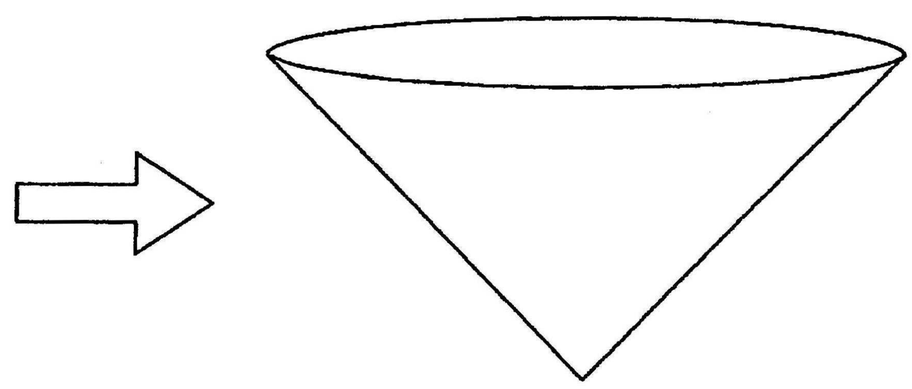


Figure 4(b)

(a) Find the length of OX. (2 marks)

(b) Find the height of the container. (4 marks)

(c) Suppose that the container is held vertically. If water of volume $150\,\text{cm}^3$ is poured into the container, will the water overflow? Explain your answer. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

SECTION B (33 marks)
Answer any THREE questions in this section and write your answers in the spaces provided.
Each question carries 11 marks.

14. In a bank, the queuing times (in minutes) of 12 customers are recorded as follows:

5.1 , 5.2 , 5.4 , 6.1 , 6.7 , 7.1 , 7.4 , 7.7 , 7.8 , 8.4 , 9.0 , 10.1

It is found that if the queuing time of a customer in the bank is less than 8 minutes, then the probability that the customer makes a complaint is $\frac{1}{6}$. Otherwise, the probability that the customer makes a complaint is $\frac{1}{3}$.

(a) If a customer is randomly selected from the 12 customers, find the probability that the selected customer does not make a complaint. (3 marks)

(b) Two customers are now randomly selected from the 12 customers.

(i) If the queuing time of one of the selected customers is less than 8 minutes and the queuing time of the other customer is not less than 8 minutes, find the probability that both of them do not make complaints.

(ii) Find the probability that the queuing times of both of the selected customers are not less than 8 minutes and both of them do not make complaints.

(iii) Is the probability of not making complaints by the two selected customers greater than the probability of making complaints by both of them? Explain your answer.
(8 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

15. Figure 5 shows a sequence of tables filled with integers. The 1st table consists of 1 row and 1 column and 1 is assigned to the cell of the 1st table. For any integer $n > 1$, the nth table consists of n rows and n columns and the integers in the cells of the nth table satisfy the following conditions:

(1) The integer in the cell at the top left corner is n.

(2) In each row, the integer in the cell of the $(r+1)$ th column is greater than that of the rth column by 1, where $1 \leq r \leq n-1$.

(3) In each column, the integer in the cell of the $(r+1)$ th row is greater than that of the rth row by 1, where $1 \leq r \leq n-1$.

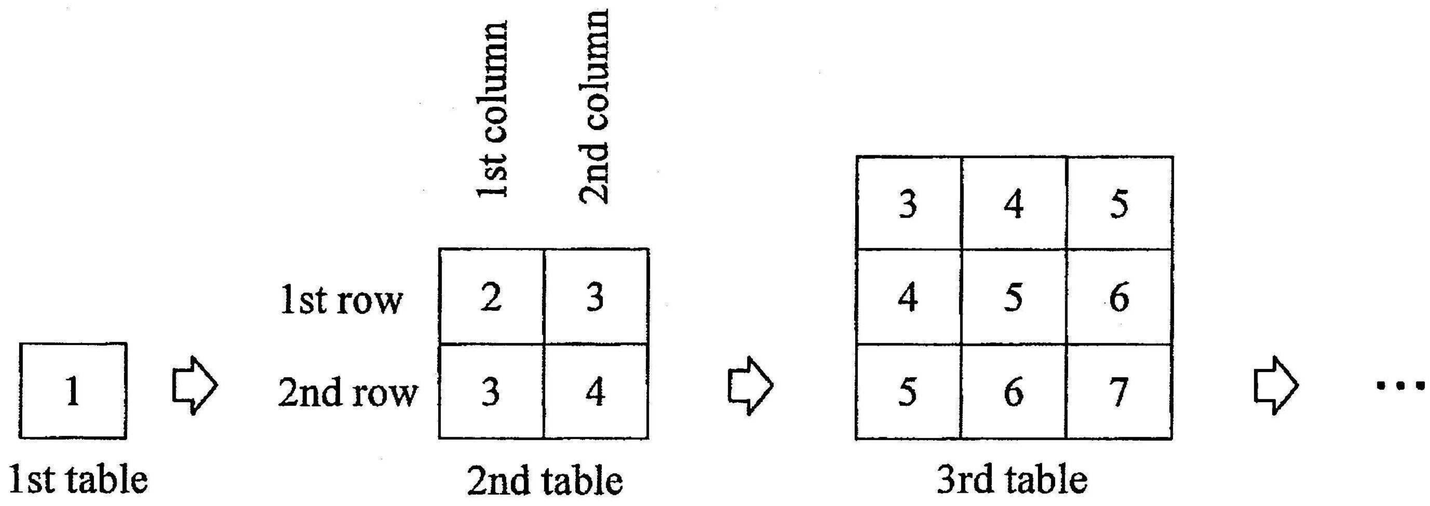


Figure 5

(a) Construct and complete the 4th table. (2 marks)

(b) Find the sum of all integers in the 1st row of the 99th table. (2 marks)

(c) Find the sum of all integers in the 99th table. (3 marks)

(d) Is there an odd number k such that the sum of all integers in the kth table is an even number? Explain your answer. (4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

16. In Figure 6, ΔPQR is an isosceles triangle with $PQ = PR$. It is given that S is a point lying on QR and the orthocentre of ΔPQR lies on PS. A rectangular coordinate system is introduced in Figure 6 so that the coordinates of P and Q are $(16, 80)$ and $(-32, -48)$ respectively. It is given that QR is parallel to the x-axis.

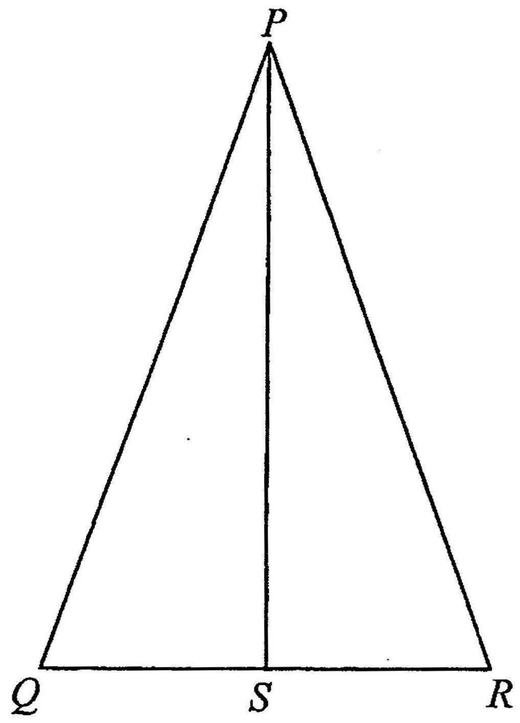


Figure 6

(a) Find the equation of the perpendicular bisector of PR. (4 marks)

(b) Find the coordinates of the circumcentre of ΔPQR. (3 marks)

(c) Let C be the circle which passes through P, Q and R.

(i) Find the equation of C.

(ii) Are the centre of C and the in-centre of ΔPQR the same point? Explain your answer. (4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Go on to the next page

Page total

17. In Figure 7(a), ABC is a thin triangular metal sheet. D and E are points lying on AB and AC respectively such that DE is parallel to BC and the distance between DE and BC is 4 cm. It is found that $AB = 20$ cm, $AC = 30$ cm and $\angle BAC = 56°$.

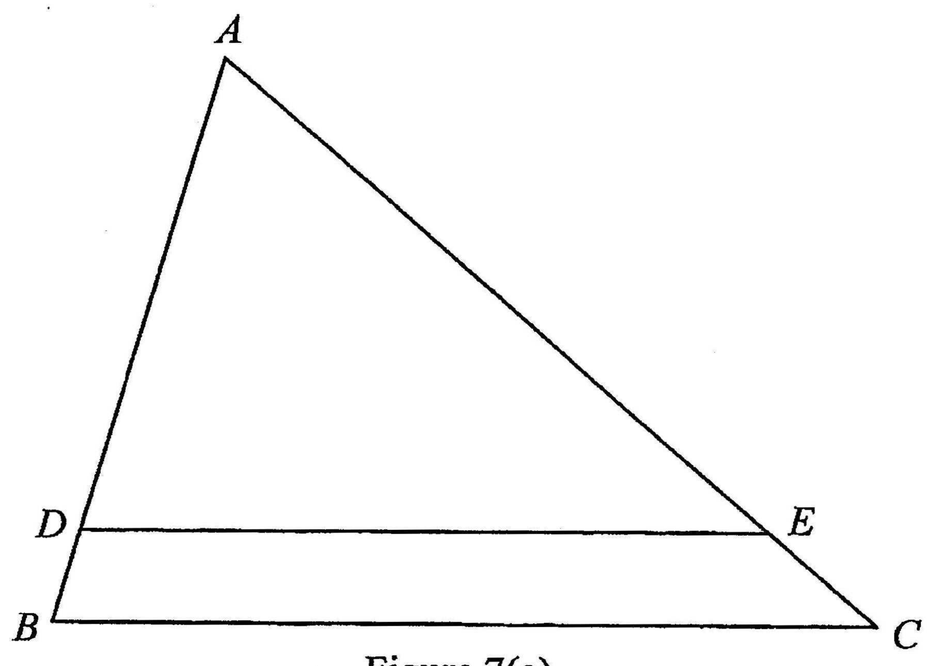


Figure 7(a)

(a) Find

(i) the length of BC,

(ii) $\angle ACB$,

(iii) the perpendicular distance from A to DE,

(iv) the length of DE.

(8 marks)

(b) The thin triangular metal sheet in Figure 7(a) is cut along DE. The metal sheet ADE is held with DE lying on the horizontal ground as shown in Figure 7(b). It is given that P is the projection of A on the horizontal ground and the area of ΔPDE is $120\ \text{cm}^2$.

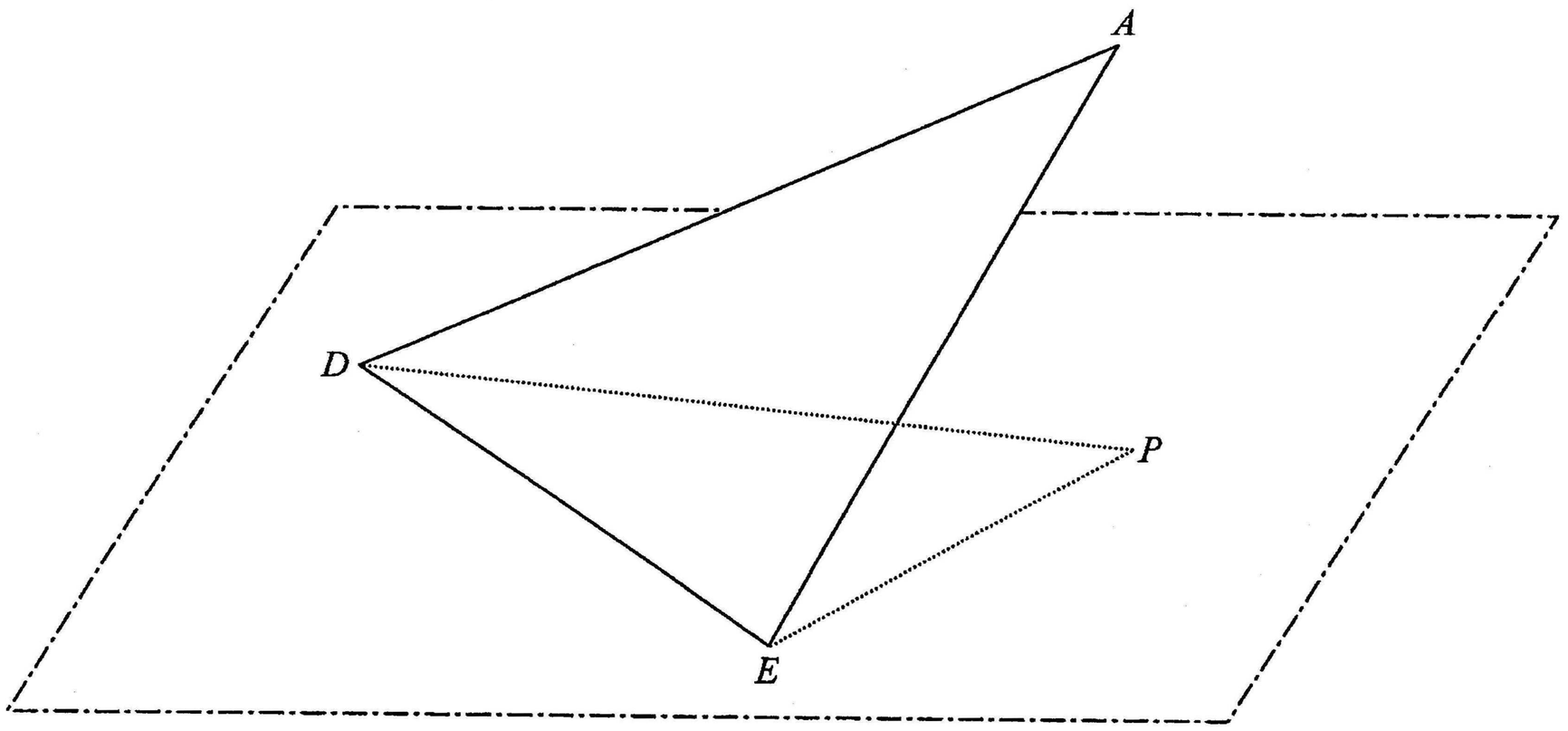


Figure 7(b)

Find

(i) the angle between the metal sheet ADE and the horizontal ground,

(ii) the shortest distance from A to the horizontal ground.

(3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Page total

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

END OF PAPER

Answers written in the margins will not be marked.

Page total